How little does learning direct deduction actually require?

All 51 cells · N=48, degree 5, 4 colours, fall ensemble, leak-free (--shuffle-adj)

THE QUESTION

Not how well the model solves direct deduction — it already does, leak-free — but how little machinery that takes. Every cell is the L=12/ff=1024 model with something REMOVED: layers, heads, the MLP's nonlinearity, its training, or the MLP write itself.

THE HEADLINE

There is no knee. One seed at L=4 never crosses 50%, but L=3 crosses on all three and ties L=4 on mean solve rate (0.3102 vs 0.3084) — so the censoring at L=4 is the seed term, not 4 layers. Deduction degrades GRADUALLY with depth and is gone by L=1.

BEFORE AND AFTER, at the shallowest depth that still deduces

BEFORE: L=6, 4 heads, trained ReLU MLP — rho_50 = 0.592. Removable for free at L=6 (rho_50 rises by less than s*): nothing. Not removable: linear (+0.173, 1 of 3 seeds censored), rf_mlp (+0.171, 1 of 3 seeds censored), no_mlp (+0.171, 1 of 3 seeds censored), nhead=1 (censored 3/3), rf_all (censored 3/3). rf_all escapes the usual 'did not train' hedge: freezing all but the readout leaves a CONVEX objective in the 14,906 free parameters, and its loss sits at 3.60/3.60/3.59, slope +0.00002 — CONVERGED, not under-trained. Frozen features with a linear readout do none of DD (mean solve 0.0000, 3/3): architecture, not budget.

WHAT EACH REMOVAL COSTS, IN LAYERS

Each stripped L=6 cell, matched to the depth rung it scores like: linear ~ L=3 (0.3146 vs 0.3102); rf_mlp ~ L=3 (0.3104 vs 0.3102); no_mlp ~ L=4 (0.3039 vs 0.3084); nhead=1 ~ L=1 (0.0270 vs 0.0211). THE THREE MLP CUTS COST THE SAME: mean solve 0.3146 / 0.3104 / 0.3039, a spread of 0.0107 against the L=6 hub's own across-seed spread of 0.0141. Deleting the MLP entirely costs what removing only its nonlinearity costs, and what freezing only its first matrix costs — the MLP is ALL-OR-NOTHING, only its LEARNED nonlinearity buys anything, and it is worth about half the depth (L=3/4, two rungs the ladder itself cannot separate).

HOW TO READ EVERY NUMBER

rho_50 is the hint density where the solve rate crosses 50% — LOWER IS BETTER. It censors when a cell never crosses, so mean solve rate over the 14-point grid is quoted beside it. s* is the largest across-seed spread on any complete rung in EITHER job — pooled, so moving the arch axis cannot buy a smaller bar: 0.148 in rho_50, 0.292 in mean solve.

Taken away — depth: L = 12, 8, 6, 4, 3, 2, 1. Architecture at L=4: linear, rf_mlp, rf_all, no_mlp; heads 4->1 measured. Architecture at L=6: linear, rf_mlp, rf_all, no_mlp; heads 4->1 measured. Three seeds on every rung. Every cell: d_model 256, ff 1024, pre-norm, constant lr 1e-3, batch 32, 240k instances x 50 epochs = 375k gradient steps — finding 44's budget and optimizer, unchanged on purpose. Same 14-point x 400-instance grid and the same instances as the 11-02 scan, so the curves are paired point by point.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-57	N0 N1 ... N47
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*

SOLVED

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image:
node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 401 TOKENS

HINTS	BEGIN_HINTS N0 COLOR_0 N1 COLOR_2 N2 COLOR_3 N10 COLOR_1 N15 COLOR_1 N17 COLOR_3 ... (37 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N8 N38 N34 N17 N1 N13 N27 N34 N28 N10 ... (289 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 N34 COLOR_1 N44 ... (75 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 13 rounds, 30 deductions,
60 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000018, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

18 hints of 48 nodes, then 13 rounds of simultaneous deduction. Hints: N0=COLOR_0 N1=COLOR_2 N2=COLOR_3 N10=COLOR_1 N15=COLOR_1 N17=COLOR_3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND      N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 ...
  6 deduced
BEGIN_ROUND      N42 COLOR_3 N45 COLOR_1
  2 deduced
BEGIN_ROUND      N13 COLOR_0 N20 COLOR_0 N37 COLOR_2
  3 deduced
BEGIN_ROUND      N21 COLOR_2 N24 COLOR_3 N31 COLOR_2
  3 deduced
... 9 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N9, neighbours N25 N10 N29 N31 N18

their colours known before this round:

```
[0, 1, 2]  (3 = q-1 of the 4)

(3 known neighbours, 3 DISTINCT colours)

still unknown: N25 N31
```

only colour left: 3

```
so the trace emits  N9 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

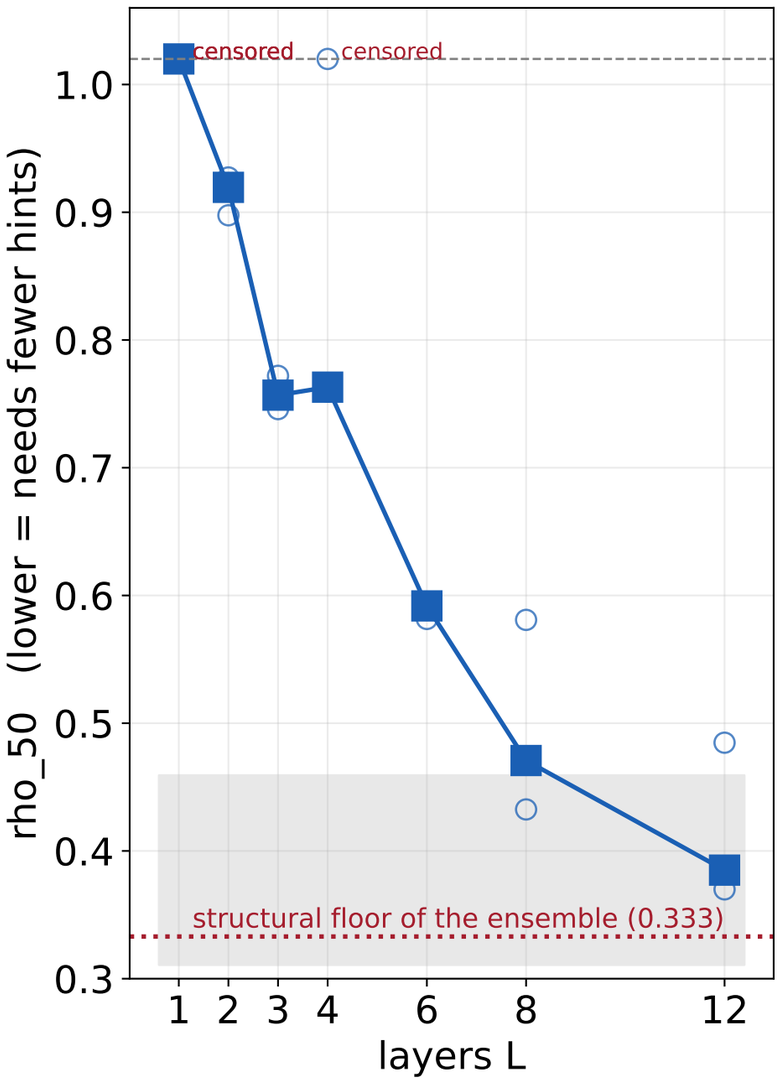
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

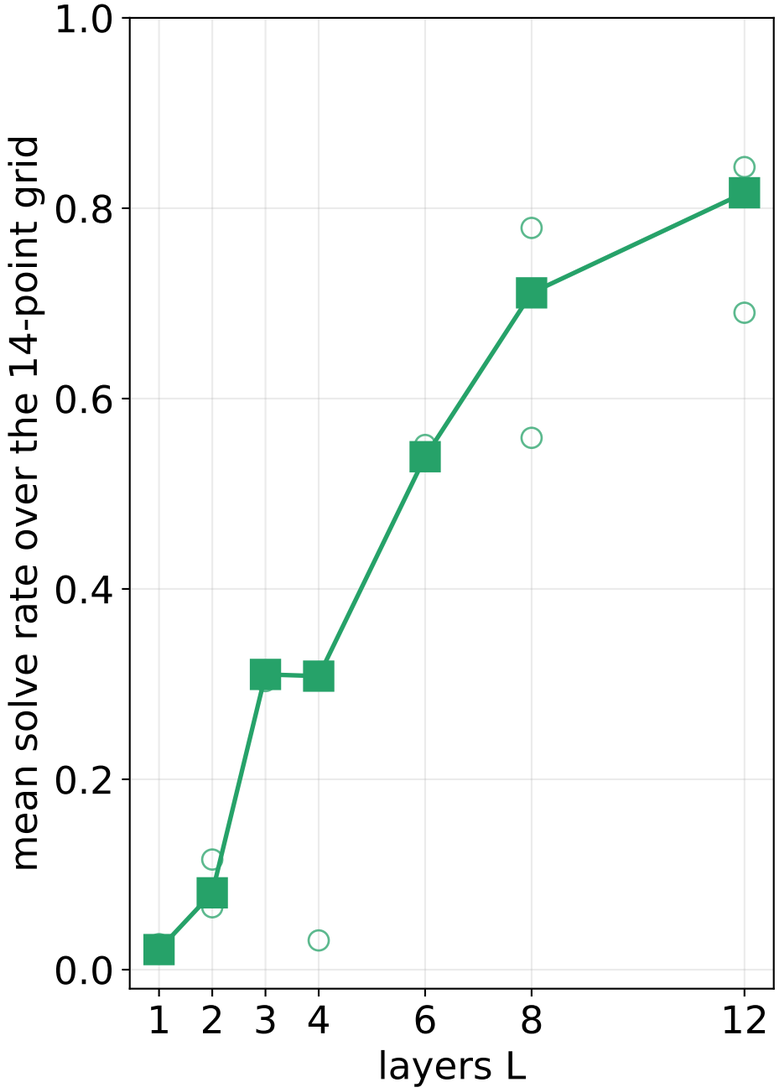
N=48, degree 5, 4 colours, --shuffle-adj --fall: the task every cell in this deck is trained and scored on. This is what the models have to produce token by token, with no scratchpad and no tool — the question of the deck is how much network that needs.

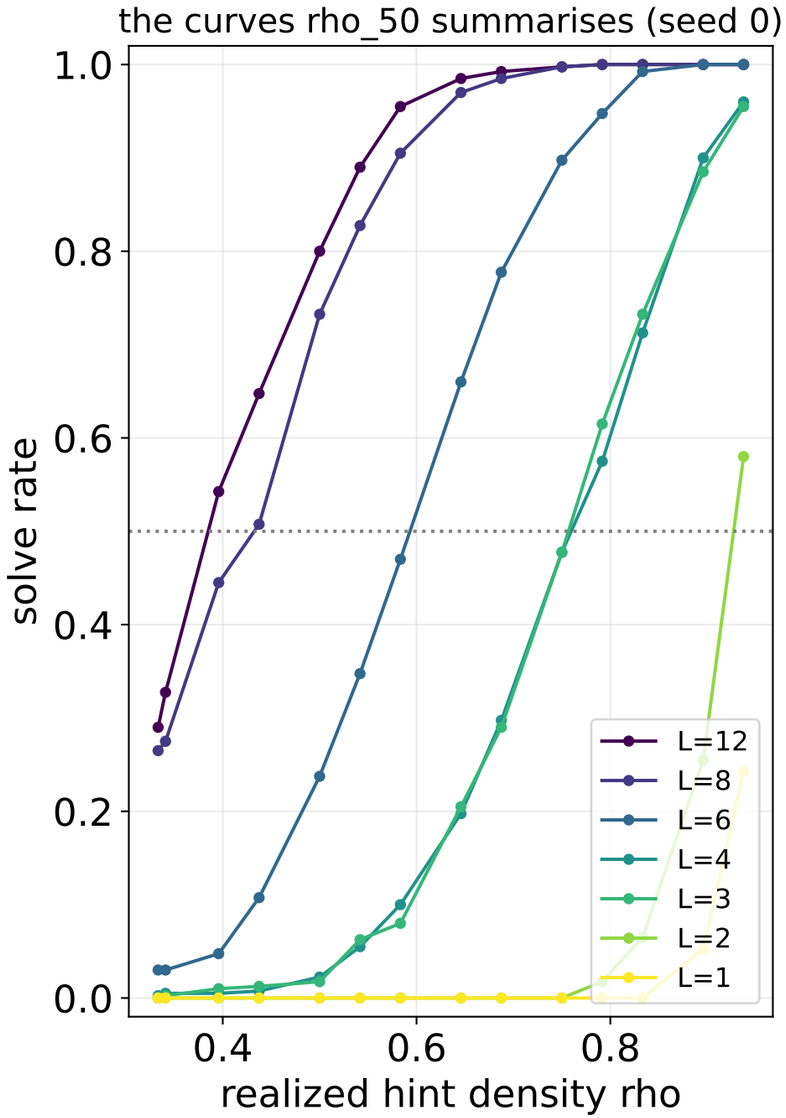
T=13 rounds, the 90th percentile of 40 draws from this config (range 1-16) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

Walking the depth ladder down

nhead 4, full MLP, three seeds per rung; lower rho_50 is better



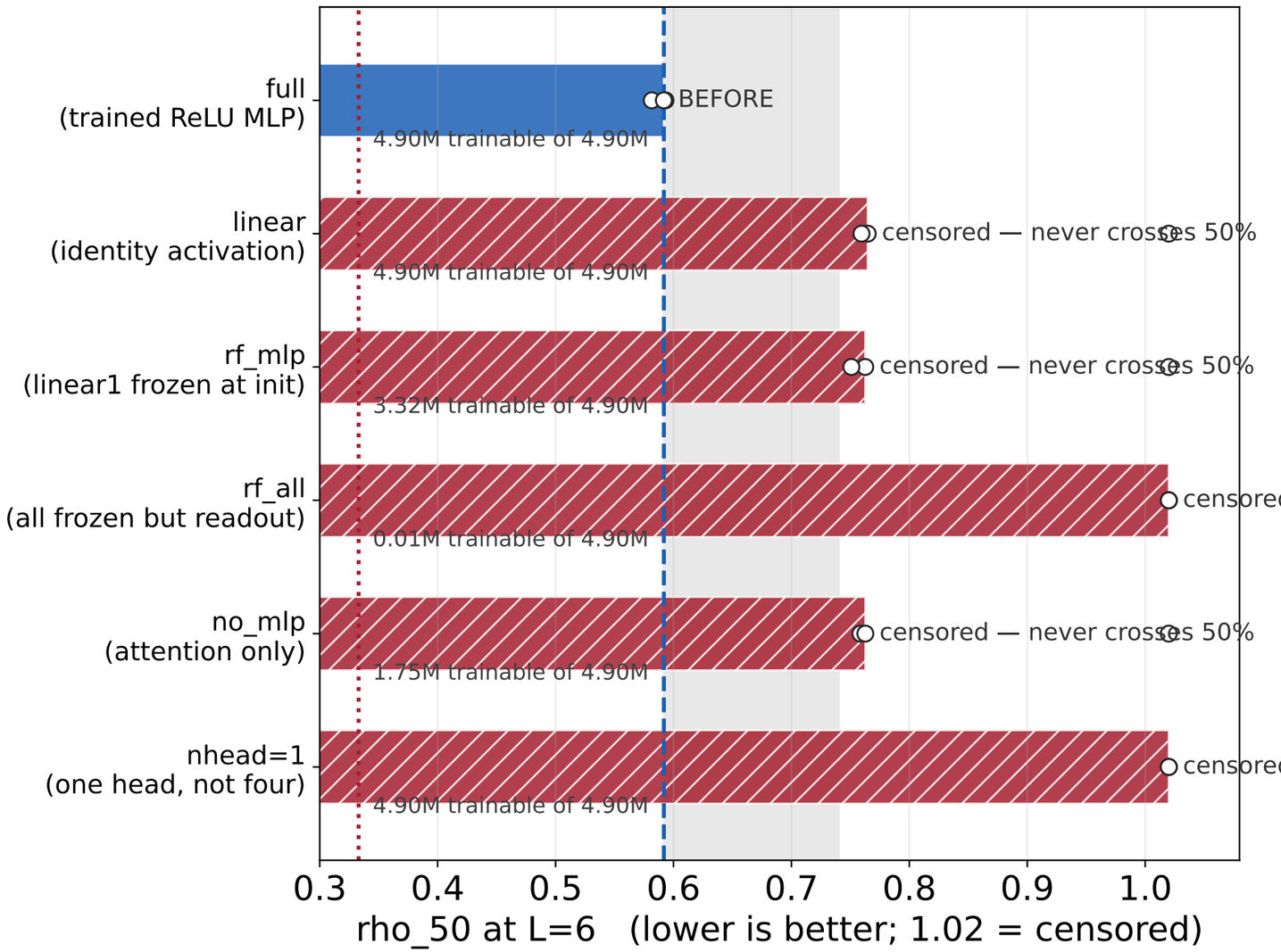




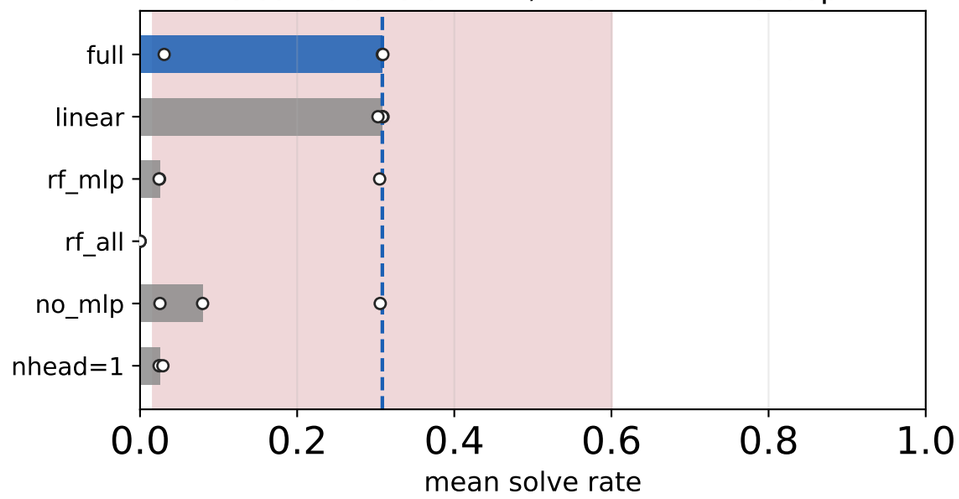
Open circles are the three seeds, filled squares the median. The grey band on the left is +/- s*/2 about the DEEPEST rung measured — the model being shrunk from — so a rung whose median sits inside it has lost nothing readable by being smaller. Points drawn at 1.02 are cells that never crossed a 50% solve rate anywhere in the grid, so their rho_50 is a bound and not a value — read those rungs on the right-hand panel instead.

Stripping the machinery: rho_50 before and after, at L=6

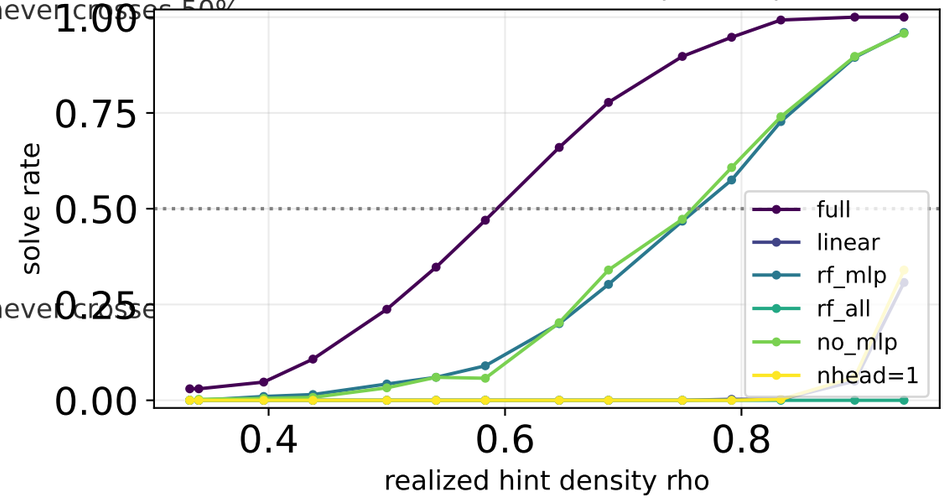
same 375k steps, same seeds, same initialization — only what is trainable changes



the same axis at L=4, the BREAKING depth



L=6 transition curves (seed 0)



The dashed line is the unmodified L=6 model and the grey band is the pooled s*, the largest across-seed spread on any complete rung in either job; a bar ending inside it costs nothing readable. The arch axis MOVED from L=4 to L=6 because L=4 is where deduction breaks: in the small panel the red band (s*) is as wide as the hub's whole score, so every bar there ties with every other and with a model that learned nothing. s* is pooled over both depths, so the move buys no slack. Every mode starts from the same initialization at a given seed.

Every rung

N=48 fall, --shuffle-adj, d_model 256, ff 1024, 375k steps; commit 1f94ad2

L	head	arch	trainable	seeds	rho_50 per seed	median	spread	mean solve	spread	floor	train NLL	escape
12	4	full	9.64M	3	0.385/0.370/0.485	0.385	0.115	0.8163	0.1530	0.290	1.495	3/3
8	4	full	6.48M	3	0.432/0.581/0.471	0.471	0.148	0.7111	0.2205	0.138	1.505	3/3
6	4	full	4.90M	3	0.593/0.582/0.592	0.592	0.012	0.5389	0.0141	0.030	1.528	3/3
4	4	full	3.32M	3	0.760/ >0.95/0.763	censored	--	0.3084	0.2784	0.003	1.573	3/3
3	4	full	2.53M	3	0.757/0.772/0.746	0.757	0.026	0.3102	0.0071	0.000	1.606	3/3
2	4	full	1.74M	3	0.927/0.919/0.898	0.919	0.030	0.0807	0.0502	0.000	1.688	0/3
1	4	full	0.95M	3	>0.95/ >0.95/ >0.95	censored	--	0.0211	0.0062	0.000	2.452	0/3
4	4	linear	3.32M	3	0.772/0.761/0.765	0.765	0.010	0.3086	0.0064	0.000	1.649	1/3
4	4	rf_mlp	2.27M	3	>0.95/0.769/ >0.95	censored	--	0.0248	0.2807	0.000	1.670	0/3
4	4	rf_all	0.01M	3	>0.95/ >0.95/ >0.95	censored	--	0.0000	0.0000	0.000	3.630	0/3
4	4	no_mlp	1.22M	3	0.769/0.921/ >0.95	censored	--	0.0796	0.2804	0.000	1.702	0/3
4	1	full	3.32M	3	>0.95/ >0.95/ >0.95	censored	--	0.0255	0.0048	0.000	1.672	0/3
6	4	linear	4.90M	3	>0.95/0.765/0.759	censored	--	0.3146	0.2918	0.000	1.588	2/3
6	4	rf_mlp	3.32M	3	0.763/0.751/ >0.95	censored	--	0.3104	0.2693	0.000	1.589	2/3
6	4	rf_all	0.01M	3	>0.95/ >0.95/ >0.95	censored	--	0.0000	0.0000	0.000	3.599	0/3
6	4	no_mlp	1.75M	3	0.758/ >0.95/0.763	censored	--	0.3039	0.2671	0.000	1.631	2/3
6	1	full	4.90M	3	>0.95/ >0.95/ >0.95	censored	--	0.0270	0.0032	0.000	1.650	0/3

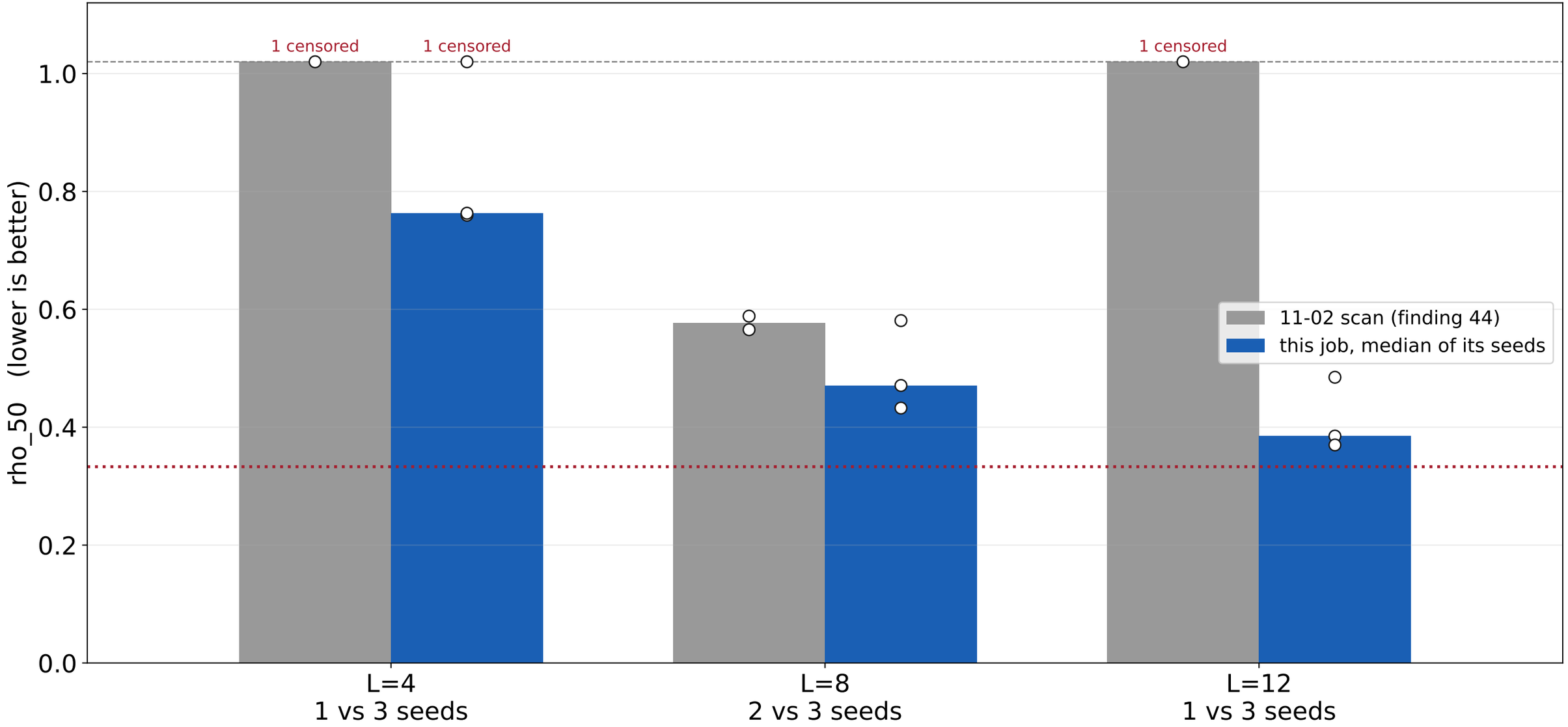
pooled yardstick s* = 0.1484 in rho_50, 0.2918 in mean solve — the largest across-seed spread on any complete rung above, taken over BOTH jobs

SAME TRAINABLE BUDGET, SPENT TWO WAYS — each stripped L=6 rung against the depth rung nearest it in trainable count:
no_mlp L=6 1.746M vs full L=2 1.741M (within 6k): mean solve 0.3039 vs 0.0807 -> stripped WINS
rf_mlp L=6 3.321M vs full L=4 3.320M (within 1k): mean solve 0.3104 vs 0.3084 -> tie

'escape' counts the seeds whose final training NLL got below 1.64, where finding 44's L=4 cell plateaued for its whole run at this budget. A rung with 0 of 3 escapes did not TRAIN here; that is not the same as being unable to represent deduction, and it is not written up as such. 'floor' is the solve rate at rho = 0.333, the hardest bin in the grid.

Against the previous best on this exact eval

finding 44's N=48 fall cells (11-02): same 375k steps, same d_model and ff, same 14-point grid, same instances



This is a REPLICATION panel, not an improvement claim: the configuration is identical and the only change is three seeds instead of one. The white dots show how much of the apparent difference is the seed. The point of this job is smallness, so a rung that merely matches its 11-02 twin at a third of the depth is the result being looked for.

Controls, and what this measurement cannot carry

all pre-registered in the sbatch header before submission

LEAK — the control that matters most in exactly this regime

Every cell trains AND evaluates with --shuffle-adj, read back out of each written config.json rather than trusted from the command line: [('True', 'True')] over all 714 rows. Findings 16, 17 and 23 are why. A minimal architecture that scores well on the leaky task has told us nothing, and the smaller the model the more tempting the decoding shortcut becomes.

GRAMMAR vs DEDUCTION

Over the 254 bins with solve rate above 0.1: worst truncation 0.0200 against a 0.01 bar, worst well-formed rate 0.9500 against 0.99. 22 bins breach one of them — small breaches, but reported rather than rounded away. They are concentrated at the low-density end, where traces are longest, and they are far too small to carry any gap on the other pages.

TRAINABILITY IS NOT CAPACITY

These rungs had NO seed get below finding 44's 1.64 training-NLL plateau: L=1 h=4 full, L=2 h=4 full, L=4 h=1 full, L=4 h=4 no_mlp, L=4 h=4 rf_all, L=4 h=4 rf_mlp, L=6 h=1 full, L=6 h=4 rf_all. They are reported as 'did not train at 375k steps', NOT as 'cannot represent deduction'. Separating the two needs more steps or a better optimizer, neither of which is in this job. PARTIAL, and it matters more: L=4 h=4 linear seed(s) 1 at 1.669, 2 at 1.649; L=6 h=4 linear seed(s) 0 at 1.667 — and it is the censored one; L=6 h=4 no_mlp seed(s) 1 at 1.688 — and it is the censored one; L=6 h=4 rf_mlp seed(s) 2 at 1.642 — and it is the censored one. Where the stuck seed is also the censored one, part of that rung's cost is trainability, not representational power, and the quoted gap is the cost on the seeds that did train.

WHAT THIS DECK CANNOT CARRY

51 of 51 cells. Three seeds is enough to put a spread on a rung and not enough for a confidence interval on the spread itself. rho_50 is one number summarising 14 binomial bins at n=400, and it censors; where it does, the mean solve rate carries the comparison and the page says so. Nothing here is causal — a small model that solves the task shows the task does not need more machinery, not that this machinery is what DD looks like.

The config.json assertion runs inside each task, between training and evaluation, so a cell whose arch_mode, shuffle_adj, fall, shape or max_len margin is not what was asked for fails loudly instead of producing a plausible number — finding 43's failure mode, on a knob that is exactly as silent.